ZOO519P – PHYSIOLOGY

Practical Exam Notes

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MCQs (Multiple Choice Questions)

1. The total volume of fluid under the coverslip in one ruled area of a hemocytometer is **0.1mm^3**.
2. What dilution factor is commonly used for RBC counting? **1:200**
3. Counting area of WBC is larger than RBC counting area because **WBCs ARE FEWER IN NUMBER**.
4. The main reason for slow charging of hemocytometer is to **ENSURE EVEN DISTRIBUTION**.
5. Normal blood sugar range in fasting condition is **70 – 100** mg/dl.
6. Which hormone regulates blood sugar levels by lowering them? **INSULIN**
7. In the GTT (Glucose Tolerance Test), blood sugar levels are measured at intervals after consuming glucose. What is the critical time interval for diabetes diagnosis? **2 HOURS**
8. Which organ releases glucagon during hypoglycemia? **PANCREAS**
9. Which method of hemoglobin measurement is considered the gold standard? **CYANMETHEMOGLOBIN METHOD**
10. In the Sahli's method, what is used to convert hemoglobin to acid hematin? **HYDROCHLORIC ACID**
11. What color is the solution in the cyanmethemoglobin method? **YELLOW**
12. Which of the following can interfere with hemoglobin measurement? **ALL OF THE GIVEN**
13. What is the hemoglobin content in a healthy male (g/dL)? **13.5 – 17.5**
14. What device is used to measure blood pressure? **SPHYGMOMANOMETER**
15. When cuff pressure is above systolic pressure, what happens to blood flow sounds? **THEY DISAPPEAR**
16. Resting membrane potential is mainly maintained by which mechanism? **SODIUM-POTASSIUM PUMP**
17. Which ion is responsible for depolarization during nerve impulse? **SODIUM**
18. What is the typical value of resting membrane potential in a neuron? **-70mV**
19. Which hormone is secreted during the luteal phase of the estrous cycle? **PROGESTERONE**
20. What is the role of prostaglandin F2α in the estrous cycle? **CAUSES LUTEOLYSIS**
21. Which phase of the estrous cycle is characterized by ovulation? **ESTRUS**
22. Which phase in the cardiac cycle is represented by the 'lub' sound? **VENTRICULAR SYSTOLE**
23. What is the term for measurement of pressure in the arteries when the heart is at rest between beats? **DIASTOLIC PRESSURE**
24. At which wavelength, cyanmethemoglobin has a broad absorbance peak? **540 nm**
25. A kymograph is an instrument used for recording variations in **BOTH BLOOD PRESSURE AND TENSION OF MUSCLES**.

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26. Which of the following compound is used in hemoglobinometery? **CYANMETHEMOGLOBIN**
27. What is the location of the pacemaker region in the heart of frog and turtle? **SINUS VENOSUS**
28. The blood flow sounds will continue until the cuff's pressure **FALLS BELOW** the arterial diastolic pressure.
29. How many time carbon monoxide has more affinity for hemoglobin than oxygen? **200**
30. Why should one avoid touching a nerve with any metal object? **IT WILL OVER EXCITE THE NERVE**
31. The cells responsible for spontaneous action potential production in the heart are often referred to as **PACEMAKER CELLS**.
32. Red Blood Cells are also called **ERYTHROCYTES**.
33. Which of the following factors can contribute to high blood pressure? **OBESITY**
34. Which of the following objective lens is used to observe a large central square in the hemocytometer? **10X**
35. **POTASSIUM CYANIDE** converts hemoglobin to methemoglobin.
36. What is the first step in nerve-muscle preparation nerve-muscle? **FIXING THE TOAD ON THE FROG BOARD**
37. It's normal for systolic blood pressure to rise to between **160** and **220** mm Hg during exercise.
38. All forms of hemoglobin are converted to cyanmethemoglobin except **SULFHEMOGLOBIN**.
39. The electrical impulse that regulates the heartbeat originate in the **SINOATRIAL NODE**.
40. Which of the following is not the function of Ringer's solution? **OXYGEN TRANSPORT**
41. In a sphygmomanometer, which component is used to release pressure from the cuff? **VALVE**
42. What should be done if the cells touch the 4 perimeter sides of a corner square? **COUNT CELLS ON 2 SIDES**
43. Kymograph was invented by German physiologist Carl Ludwig in **1840s**.
44. Brunner's gland is located in **SUBMUCOSA OF DUODENUM**.
45. Bile pigments are derived from **HEMOGLOBIN**.
46. Villus has a covering of **SIMPLE COLUMNAR EPITHELIUM**.
47. Amylase is secreted by **SALIVARY GLANDS** or **PANCREASE**.
48. There are **TWO** adrenal glands present in humans.
49. Cardiac output of resting adult human is **5 L/min**.

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SHORT QUESTIONS

1. What are the key components of DO meter? How they are used to measure oxygen concentration?

A DO (Dissolved Oxygen) meter measures the amount of dissolved oxygen in water, usually in mg/L (ppm) or % saturation.

Key Components

1. ***DO Probe (Sensor):*** Detects dissolved oxygen in water.
2. ***Membrane:*** Allows only oxygen to pass into the probe.
3. ***Electrolyte Solution:*** Conducts ions and supports the electrochemical reaction.
4. ***Anode and Cathode (Electrodes):*** Generate an electrical current proportional to the oxygen concentration.
5. ***Temperature Sensor:*** Compensates for temperature changes that affect oxygen solubility.
6. ***Display Unit:*** Converts the electrical signal into a digital DO reading.

Working Principle

- The probe is immersed in the water sample.
- Dissolved oxygen passes through the membrane into the probe.
- Oxygen reacts at the electrodes, producing an electrical current.
- The magnitude of the current is proportional to the dissolved oxygen concentration.
- The meter processes the signal, applies temperature compensation, and displays the dissolved oxygen value in mg/L or % saturation.

2. Write formula for RBC count in hemocytometer.

Formula for RBC count using a hemocytometer:

$$RBC\ Count\ (cells/mm^3) = \frac{number\ of\ cells\ counted\ x\ dilution\ factor}{area\ counted\ x\ depth}$$

For the standard RBC counting method:

- Dilution factor = 200
- Area counted = 0.2 mm²
- Depth = 0.1 mm

Therefore,

$$RBC\ Count = Number\ of\ cells\ counted\ x\ 10{,}000\ cells/mm^3$$

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3. How do researchers investigate specific behavior?

Researchers investigate specific behavior using scientific methods to collect accurate and reliable data. The main methods include:

- ***Observation:*** Researchers observe and record behavior in either natural settings (naturalistic observation) or controlled laboratory environments.
- ***Experiments:*** Variables are manipulated under controlled conditions to determine the cause-and-effect relationship between different factors and behavior.
- ***Surveys and Questionnaires:*** Information is collected from a large number of participants regarding their behaviors, attitudes, opinions, or experiences.
- ***Interviews:*** Researchers ask structured or unstructured questions to obtain detailed information about an individual's behavior and thoughts.
- ***Case Studies:*** An in-depth study of a single individual, group, or event is conducted to understand specific behaviors in detail.
- ***Psychological Tests:*** Standardized tests are used to assess intelligence, personality, memory, or other behavioral characteristics.

4. Systolic pressure increases with exercise. Do you agree with this statement? If yes, then clarify why.

Yes, the statement is correct. Systolic pressure increases during exercise because the heart needs to pump more blood to the working muscles.

During exercise, muscles require more oxygen and nutrients, so the heart increases its rate and force of contraction. This results in a greater amount of blood being pumped out of the left ventricle with each heartbeat, causing an increase in systolic blood pressure.

The increase in systolic pressure helps maintain adequate blood flow to muscles and meet the body's increased metabolic demands. After exercise, when the body's oxygen demand decreases, the heart rate and systolic pressure gradually return to normal levels.

5. Write primary functions of kymograph.

Primary functions of a kymograph:

1. ***Recording physiological movements:*** It records movements and changes in biological systems, such as muscle contractions, heartbeat, breathing, and blood pressure.
2. ***Providing graphical representation:*** It converts mechanical movements into a wave-like graphical record on a rotating drum.
3. ***Measuring time and frequency:*** It helps determine the rate, duration, and frequency of physiological events.
4. ***Studying muscle activity:*** It is used to analyze muscle contraction, relaxation, fatigue, and response to different stimuli.
5. ***Monitoring experimental changes:*** It allows researchers to observe and compare physiological responses under different experimental conditions.

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6. Write role of GnRH.

Role of GnRH (Gonadotropin-Releasing Hormone)

- GnRH is a neurohormone secreted by the hypothalamus that plays a central role in regulating the reproductive system.
- It is released in a pulsatile manner and acts on the anterior pituitary gland to stimulate the secretion of two important gonadotropins: follicle-stimulating hormone (FSH) and luteinizing hormone (LH).
- FSH and LH regulate the activity of the gonads (ovaries and testes) and control the production of gametes and sex hormones.
- In females, GnRH regulates the menstrual cycle by controlling follicular development, ovulation, and the production of estrogen and progesterone.
- In males, GnRH stimulates LH release, which promotes testosterone production, and FSH release, which supports sperm production.
- GnRH is therefore essential for sexual development, reproductive maturity, fertility, and maintenance of normal reproductive functions.

7. Write effect of Prostaglandin (PGF2a) on estrous cycle.

Effect of Prostaglandin ($PGF_2\alpha$) on Estrous Cycle

- Prostaglandin $F_2\alpha$ ($PGF_2\alpha$) is a naturally occurring reproductive hormone secreted mainly by the uterine endometrium. It plays a major role in regulating the estrous cycle by controlling the lifespan of the corpus luteum (CL).
- During the luteal phase of the estrous cycle, the corpus luteum produces progesterone, which maintains the uterus in a suitable condition for pregnancy. If pregnancy does not occur, the uterus releases $PGF_2\alpha$, which reaches the ovary and causes luteolysis (degeneration of the corpus luteum).
- Due to luteolysis, the level of progesterone decreases, removing the inhibitory effect on the hypothalamus and pituitary gland. This allows the release of reproductive hormones that stimulate the growth and maturation of ovarian follicles.
- Developing follicles produce estrogen, which causes behavioral and physical signs of estrus, such as increased sexual receptivity, changes in reproductive tract, and preparation for mating and ovulation.
- $PGF_2\alpha$ also helps regulate the timing of the next estrous cycle by initiating a new follicular phase after corpus luteum regression.
- In animal reproduction management, $PGF_2\alpha$ is widely used for estrous synchronization, allowing farmers and researchers to control the timing of heat and breeding in animals.

8. Write composition of Ringer's solution.

Composition of Ringer's Solution

Ringer's solution is an isotonic physiological solution that closely resembles the ionic composition of body fluids. It is commonly used to maintain isolated tissues and organs in physiological experiments and to replace lost body fluids. The standard composition of 1 litre of Ringer's solution is:

- ***Sodium chloride (NaCl):*** 8.6 g
- ***Potassium chloride (KCl):*** 0.3 g
- ***Calcium chloride ($CaCl_2$):*** 0.33 g
- ***Distilled water:*** To make 1000 mL

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9. Write procedure for measurement of hemoglobin using Drabkin's reagent.

Procedure for Measurement of Hemoglobin Using Drabkin's Reagent

Principle:

Hemoglobin is converted into cyanmethemoglobin by Drabkin's reagent. The intensity of the color produced is directly proportional to the amount of hemoglobin present in the blood sample. The concentration is measured using a colorimeter or spectrophotometer at 540 nm wavelength.

Requirements:

- Blood sample (anticoagulated blood)
- Drabkin's reagent
- Test tube
- Micropipette
- Colorimeter/spectrophotometer
- Cuvette

Procedure:

1. Take 5 mL of Drabkin's reagent in a clean test tube.
2. Add 20 µL (0.02 mL) of well-mixed blood sample to the reagent.
3. Mix the contents thoroughly and allow the mixture to stand for about 10–15 minutes at room temperature for complete conversion of hemoglobin into cyanmethemoglobin.
4. Transfer the solution into a cuvette.
5. Measure the absorbance using a colorimeter or spectrophotometer at 540 nm against a reagent blank.
6. Compare the absorbance value with a standard calibration curve or standard solution to determine the hemoglobin concentration.

Calculation:

$$Hemoglobin\ concentration\ \left(\frac{g}{dl}\right) = \frac{Absorbance\ of\ test}{Absorbance\ of\ standard} x\ Concentration\ of\ standard$$

10. What is auscultatory gap in blood pressure readings? How health care professionals can avoid it?

Auscultatory Gap in Blood Pressure Readings

An auscultatory gap is a temporary disappearance of Korotkoff sounds during the measurement of blood pressure by the auscultatory method. It occurs between the systolic and diastolic pressures while the cuff pressure is slowly released. During this gap, the examiner may not hear any sounds even though the cuff pressure is still above the diastolic level. If not recognized, it can lead to an incorrectly low estimation of systolic pressure or incorrectly high estimation of diastolic pressure.

How to Avoid Auscultatory Gap:

- Before measuring blood pressure, the healthcare professional should palpate the radial pulse and estimate the systolic pressure.
- The cuff should be inflated **20**–30 mmHg above the point where the radial pulse disappears.
- The cuff pressure should be released slowly while listening carefully for Korotkoff sounds.
- Proper cuff size and correct measurement technique should be used to obtain accurate readings.

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11. What is Neubauer chamber?

Neubauer Chamber

A Neubauer chamber (hemocytometer) is a specialized counting chamber used for the manual counting of cells present in a fluid sample, such as blood cells, sperm cells, and microorganisms. It consists of a thick glass slide with a central counting grid engraved on its surface. The grid contains precisely measured squares with a known area and depth, allowing the number of cells in a specific volume of fluid to be calculated.

Structure:

- It has a ruled counting area divided into large and small squares.
- The chamber depth is usually 0.1 mm when covered with a special coverslip.
- The grid allows accurate determination of cell concentration by counting cells under a microscope.

Uses:

- Counting red blood cells (RBCs) and white blood cells (WBCs) in blood samples.
- Determining sperm count in semen analysis.
- Counting cells in biological and microbiological samples.
- Estimating cell concentration in laboratory experiments.

12. Write effect of drugs on amphibian heart.

Effect of Drugs on Amphibian Heart

The amphibian heart is commonly used in physiological experiments to study the effects of different drugs on cardiac activity. Drugs can affect heart rate, force of contraction, and rhythm.

- ***Adrenaline (Epinephrine):*** Increases heart rate **(**positive chronotropic effect**)** and force of contraction (positive inotropic effect)**.** It enhances cardiac output by stimulating β_1-adrenergic receptors.
- ***Acetylcholine:*** Decreases heart rate and force of contraction. Produces a negative chronotropic and negative inotropic effect by stimulating parasympathetic activity.
- ***Atropine:*** Blocks the action of acetylcholine on muscarinic receptors. Increases heart rate by removing the inhibitory effect of the vagus nerve.
- ***Digitalis (Digoxin):*** Increases the force of cardiac contraction. At higher doses, it may cause irregular heartbeat and cardiac arrest.
- ***Potassium ions (K^+):*** High concentrations reduce heart rate and force of contraction. Excess potassium may stop the heart in diastole.
- ***Calcium ions (Ca^{2+}):*** Increase the force of contraction. High concentrations may cause the heart to stop in systole.

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13. Write down the importance of blood sugar level and also write its normal range.

Importance of Blood Sugar Level and Its Normal Range

Blood sugar (blood glucose) level refers to the amount of glucose present in the blood. Glucose is the primary source of energy for body cells, especially the brain and muscles.

Importance:

- Blood glucose provides energy required for various cellular activities and maintains normal body functions.
- It helps in the proper functioning of the brain, nervous system, and muscles.
- Maintaining normal blood sugar levels is essential for energy balance and overall metabolism.
- Blood glucose levels are regulated mainly by hormones such as insulin and glucagon.
- Measurement of blood sugar helps in the diagnosis and monitoring of metabolic disorders, especially diabetes mellitus.
- Abnormally high blood glucose (hyperglycemia) may indicate diabetes, while abnormally low glucose (hypoglycemia) can cause weakness, dizziness, and loss of consciousness.

Normal Range of Blood Glucose:

- ***Fasting blood glucose:*** 70–100 mg/dL
- ***Postprandial (2 hours after meal):*** Less than 140 mg/dL
- ***Random blood glucose:*** Approximately 70–140 mg/dL (in healthy individuals)

14. What is sinoatrial node?

Sinoatrial Node (SA Node)

The sinoatrial node (SA node) is a small group of specialized cardiac muscle cells located in the upper wall of the right atrium near the opening of the superior vena cava**.** It is known as the natural pacemaker of the heart because it initiates and regulates the heartbeat.

The SA node generates electrical impulses spontaneously due to its pacemaker activity. These impulses spread through the atrial muscles, causing atrial contraction and pushing blood into the ventricles.

The impulses from the SA node then pass to the atrioventricular (AV) node and through the conduction system of the heart, resulting in coordinated contraction of the ventricles.

The SA node controls the heart rate and rhythm and normally produces impulses at a rate of about 60–100 beats per minute in a resting adult. Proper functioning of the SA node is essential for maintaining normal cardiac activity.

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LONG QUESTIONS

1. The right and left hemisphere seem to be mirror images of each other. Do they have similar functions or not?

The left hemisphere of the brain mainly controls the functions of the right side of the body. It is associated with language-related abilities such as speech, reading, writing, and understanding words. It also plays an important role in logical thinking, analytical reasoning, mathematical skills, and problem-solving. The left hemisphere generally processes information in a sequential and organized manner.

The right hemisphere mainly controls the functions of the left side of the body. It is involved in creativity, imagination, spatial awareness, visual recognition, and understanding emotions. It helps in interpreting facial expressions, music, and non-verbal communication. The right hemisphere processes information in a more holistic and intuitive way.

Although both hemispheres have a similar structure and perform some common functions, their specialized roles are different. Both work together through the corpus callosum to coordinate body functions, behavior, and cognitive activities.

2. What do you know about sphygmomanometer? Write different steps to measure blood pressure with this apparatus.

Sphygmomanometer

A sphygmomanometer is an instrument used to measure arterial blood pressure. It measures the pressure exerted by blood against the walls of arteries and provides readings of systolic pressure (pressure during ventricular contraction) and diastolic pressure (pressure during ventricular relaxation). A sphygmomanometer consists of:

- ***Inflatable cuff:*** Wrapped around the arm to compress the brachial artery.
- ***Pressure bulb and valve:*** Used to inflate and deflate the cuff.
- ***Manometer:*** Measures cuff pressure in mmHg.
- ***Stethoscope:*** Used to hear Korotkoff sounds during blood flow.

Steps for Measurement of Blood Pressure

1. The person is allowed to sit comfortably and rest for a few minutes before measurement.
2. The cuff is wrapped around the upper arm at the level of the heart, with the lower edge placed about 2–3 cm above the elbow.
3. The brachial artery is located, and the stethoscope is placed over it.
4. The cuff is inflated using the pressure bulb until the brachial artery is compressed and blood flow stops.
5. The valve is slowly opened to release air from the cuff at a controlled rate.
6. The first appearance of Korotkoff sounds through the stethoscope indicates the systolic blood pressure.
7. The point at which the Korotkoff sounds disappear indicates the diastolic blood pressure.
8. The cuff is completely deflated, removed, and the blood pressure reading is recorded in mmHg.

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3. Briefly describe phases of Estrus cycle.

Phases of Estrous Cycle

The estrous cycle is a series of reproductive changes that occur in female animals between two successive periods of estrus (heat). It consists of four main phases:

1. **Proestrus:** Proestrus is the preparatory phase of the estrous cycle. During this phase, the corpus luteum regresses and progesterone levels decrease. Ovarian follicles begin to grow under the influence of FSH, and estrogen secretion increases. The reproductive tract prepares for mating, but the female usually does not show acceptance of the male.

2. **Estrus:** Estrus is the period of sexual receptivity or heat when the female accepts mating. It occurs due to high levels of estrogen produced by mature ovarian follicles. During this phase, signs such as increased activity, mucus discharge, and behavioral changes may be observed. Ovulation occurs at or near the end of estrus in many species.

3. **Metestrus:** Metestrus follows estrus and is the phase during which the corpus luteum develops from the ruptured follicle after ovulation. Progesterone secretion begins to increase, preparing the uterus for possible pregnancy.

4. **Diestrus:** Diestrus is the longest phase of the estrous cycle. The corpus luteum becomes fully functional and produces high levels of progesterone, which maintains the uterine environment for pregnancy. If fertilization does not occur, the uterus releases $PGF_2\alpha$, causing luteolysis and initiating a new cycle.

5. Write functions of Prostaglandin hormone.

Functions of Prostaglandin Hormone

Prostaglandins are locally acting lipid hormones produced by many tissues of the body. They regulate various physiological processes, especially in the reproductive system.

- ***Regulation of reproductive cycle:*** Prostaglandins, particularly $PGF_2\alpha$, cause regression of the corpus luteum (luteolysis) in the ovary, leading to a decrease in progesterone levels and initiation of a new estrous cycle.
- ***Induction of labor:*** Prostaglandins help in uterine contractions and cervical relaxation during parturition, facilitating childbirth.
- ***Regulation of ovulation:*** They assist in the process of follicle rupture and release of the ovum during ovulation.
- ***Control of smooth muscle activity:*** Prostaglandins influence the contraction and relaxation of smooth muscles in the uterus, gastrointestinal tract, and blood vessels.
- ***Inflammatory response:*** They play an important role in inflammation by causing redness, swelling, pain, and increased blood flow to affected tissues.
- ***Regulation of blood pressure:*** Some prostaglandins cause dilation of blood vessels and help maintain normal blood flow and blood pressure.
- ***Protection of the stomach lining:*** Certain prostaglandins promote mucus and bicarbonate secretion, protecting the gastric mucosa from acid damage.

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6. How temperature does affects oxygen consumption? Explain.

Effect of Temperature on Oxygen Consumption

Temperature has a significant effect on the rate of oxygen consumption because it influences the metabolic activity of organisms. Oxygen consumption is directly related to the rate of cellular respiration and energy production.

At higher temperatures, metabolic reactions occur faster, increasing enzyme activity and cellular respiration. As a result, the demand for oxygen increases, leading to a higher rate of oxygen consumption. However, if the temperature becomes excessively high, enzyme activity may decrease due to denaturation, which can reduce oxygen consumption.

At lower temperatures, metabolic processes slow down because enzyme activity decreases. This reduces the rate of cellular respiration and decreases oxygen consumption.

In poikilothermic (cold-blooded) animals, such as amphibians, oxygen consumption changes greatly with environmental temperature. An increase in temperature usually increases their metabolic rate and oxygen consumption, while a decrease in temperature lowers both metabolic rate and oxygen consumption. Thus, temperature is an important factor affecting respiratory activity and oxygen requirements.

7. Describe four ways of measuring hemoglobin concentration.

Methods of Measuring Hemoglobin Concentration

1. **Sahli's Acid Hematin Method:**
   In this method, hemoglobin is converted into acid hematin by adding hydrochloric acid. The brown-colored acid hematin solution is diluted with distilled water until its color matches with a standard comparator. The hemoglobin concentration is then read directly from the graduated tube and expressed as g/dL.

2. **Cyanmethemoglobin (Drabkin's) Method:**
   In this method, blood is mixed with Drabkin's reagent, which converts hemoglobin into stable cyanmethemoglobin. The intensity of the color is measured using a colorimeter or spectrophotometer at 540 nm. The absorbance value is compared with a standard to determine hemoglobin concentration.

3. **Oxyhemoglobin Method:**
   In this method, hemoglobin is converted into oxyhemoglobin by exposure to oxygen. The concentration is measured by determining the intensity of light absorption using a spectrophotometer. This method is based on the optical properties of oxyhemoglobin.

4. **Azidemethemoglobin Method:**
   In this method, hemoglobin is converted into azidemethemoglobin by reacting with sodium azide and other reagents. The resulting colored compound is measured photometrically, and the hemoglobin concentration is calculated from the absorbance value. This method is commonly used in automated blood analyzers.

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